GITAM University

Subject- Engineering Physics

Topic: Dielectric materials

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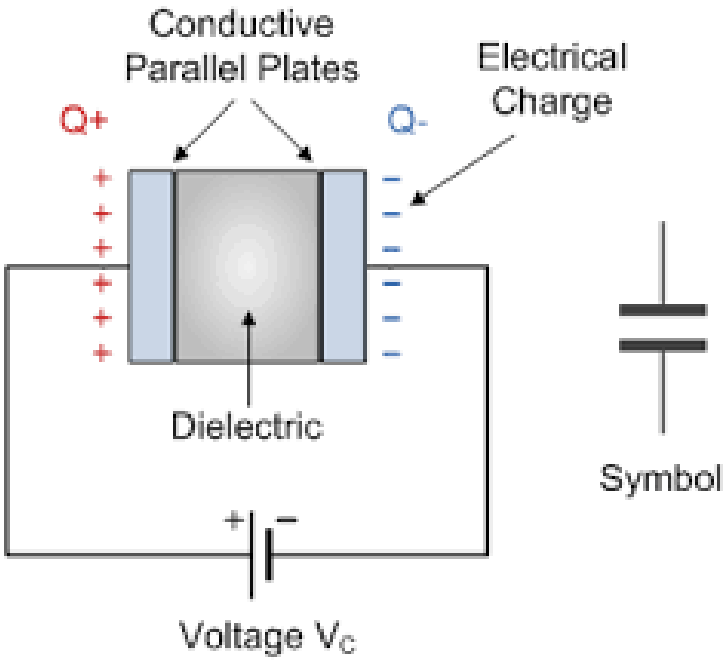


Capacitor stores electrical charge

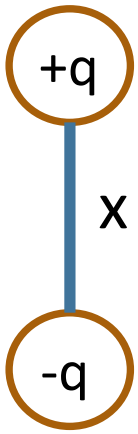
What are **Dielectric materials**

Materials having permanent electric dipole : dielectric materials

Equal negative and positive charges separated by small distance: Electric dipole

Dielectric materials are electrical insulator

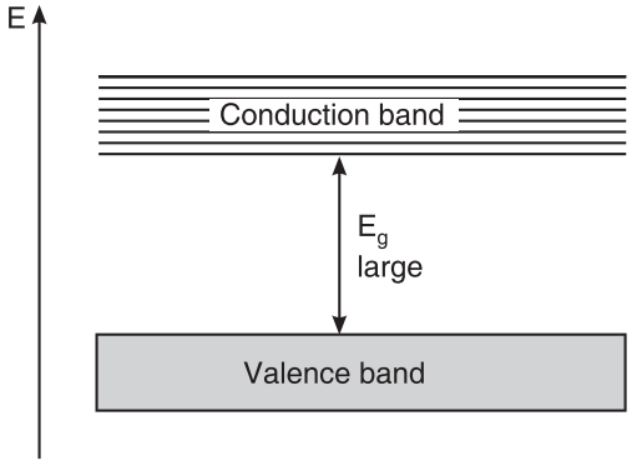
Used to store electrical energy with minimum dissipation



Dipole Moment

$$\mu_t = q \, . x$$

Unit : Coulomb. meter

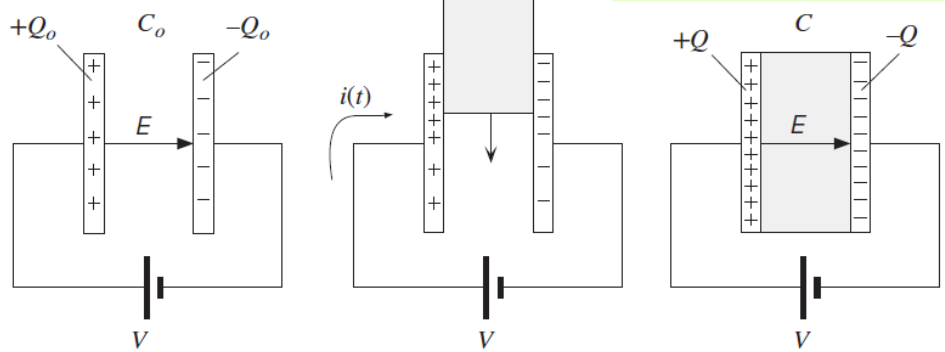


All insulators are dielectrics

Mica, Glass, Water, plastic

$$C_o = \frac{Q_o}{V} = \frac{\varepsilon_o A}{d}$$

$$C = \frac{\varepsilon_r \varepsilon_o A}{d}; \varepsilon_r = \frac{Q}{Q_o} = \frac{C}{C_o}$$



the ratio of capacitance of a capacitor containing the dielectric medium to the capacitance of the same capacitor with air as the medium : Dielectric constant

$$\kappa = \frac{C}{C_0} = \frac{\epsilon}{\epsilon_0} = \epsilon_r$$

It also a measure of amount of electrical potential energy stored in a material

ϵ_r called relative permittivity

$$C = \frac{\epsilon A}{d}$$

Energy of a capacitor, E = $\frac{1}{2}CV^2$

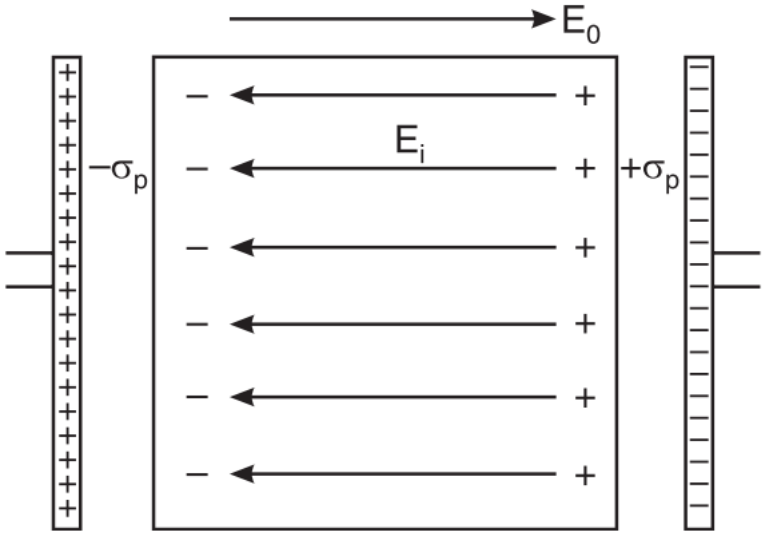
Ability to block the electric field line to pass through the material

Dielectric Polarization

When charges of opposite polarity are induced on the surfaces of a dielectric, the dielectric is said to be polarized

Intensity of polarization,

$$P = \frac{\sum d\mu}{V}, \quad d\mu = \text{dipole moment of elemental dipoles}$$



$$E_{eff} = E_0 - E_i$$

The polarized dielectric can be considered as a big dipole consisting of induced charges separated by distance d,

$$\mu = (A\sigma_P)d = \sigma_P . V$$

$$P = \sigma_P$$

$$\phi = \int D \,.\, dA = q$$

$$D = \epsilon_0 E_0 = \frac{q}{A}$$ (D = displacement vector)

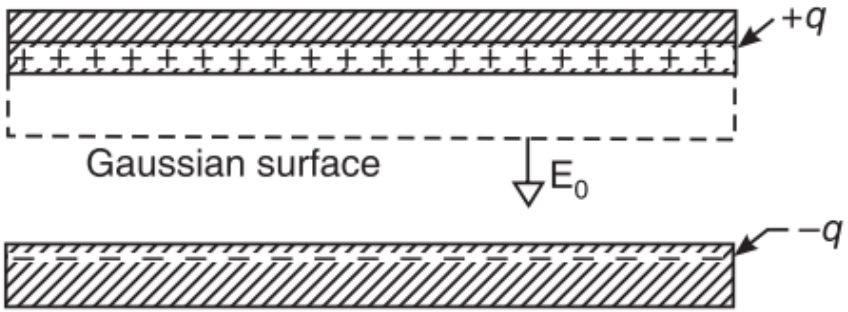
$$\rightarrow \epsilon_0 \int E_0 \,.\, dA = q$$

$$\rightarrow E_0 \, A = \frac{q}{\epsilon_0}$$

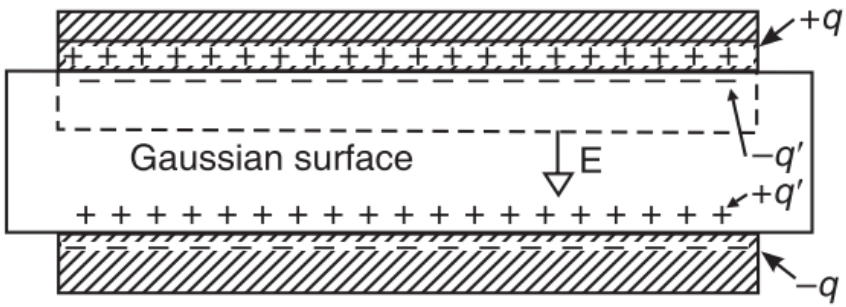
$$\rightarrow E_0 = \frac{q}{A\epsilon_0}$$

In Air

Air



Dielectric medium



In dielectric medium

$$\int E . dA = \frac{q - q'}{\epsilon_0}$$

$$\rightarrow E = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A} = \frac{E_0}{\epsilon_r}$$

$$\rightarrow \frac{q}{A\epsilon_0\epsilon_r} = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$$

$$\rightarrow q - q' = \frac{q}{\epsilon_r}$$

D, E and P relation

$$E = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$$

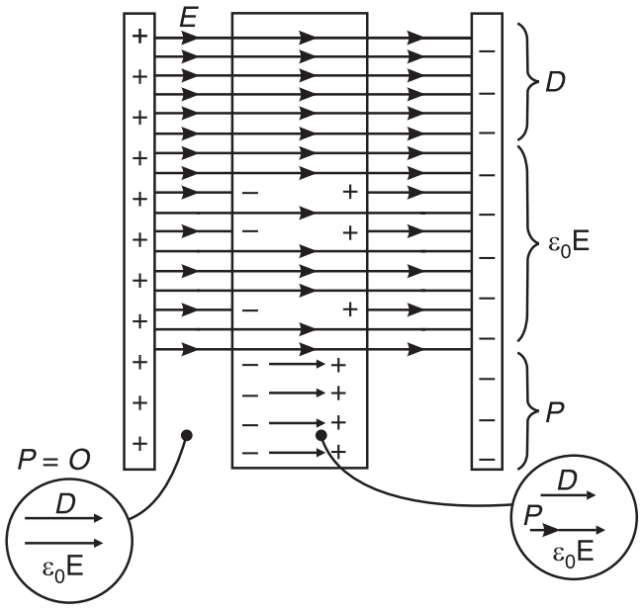
$$\sigma_P = surface\ charge\ denisty = \frac{q'}{A} = P$$

➔ $\frac{q}{\epsilon_0 A} = E + \frac{P}{\epsilon_0}$

➔ $\frac{q}{A} = \epsilon_0 E + P$

→ $\frac{q}{A}$ is called displacement vector D

➔ D = $\epsilon_0 E + P$



D related to free charges only

P related to bound charges only

E related to both

Relative permittivity with electrical susceptivity

Dielectric susceptibility

$$P = \chi \epsilon_0 E$$

χ is dielectric susceptibility
Indicates the ease with which a dielectric material can be influenced by an external electric field

D = $\epsilon_0 E + P$

→ $D = \epsilon_0 E + \chi \epsilon_0 E$ → P = $\chi \epsilon_0 E$

→ D = $(1 + \chi)\epsilon_0 E = \epsilon_0 \epsilon_r E = \epsilon E$

→ $\epsilon_r = 1 + \chi$

Relation between permittivity and susceptibility

Polarization with electric field

$$q - q' = \frac{q}{\epsilon_r}$$

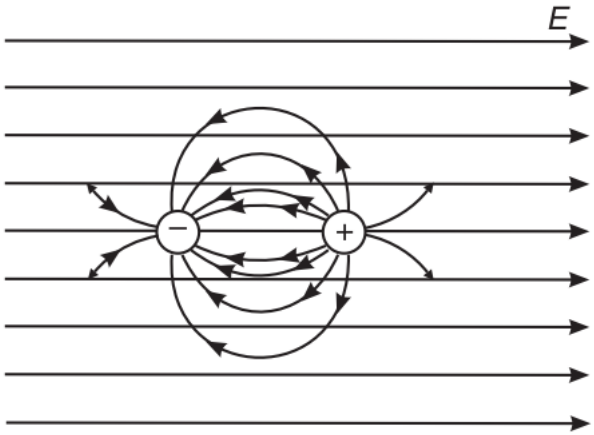
➔ $q' = q\left(1 - \frac{1}{\epsilon_r}\right)$ we know, P = $\sigma_P = \frac{q'}{A}$

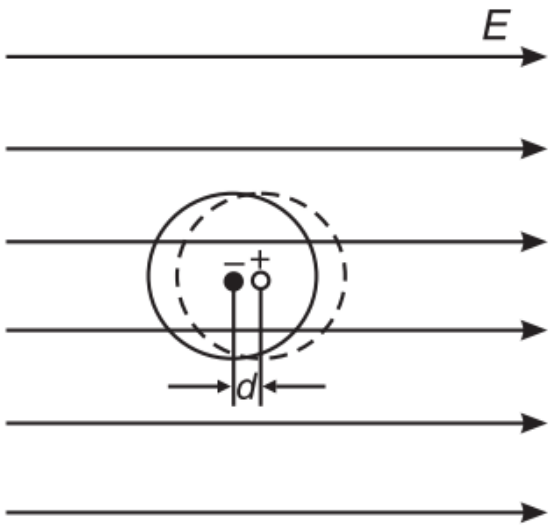
➔ $P = \frac{q}{A}\left(1 - \frac{1}{\epsilon_r}\right) = \frac{q}{\epsilon_r A}(\epsilon_r - 1) = \frac{D}{\epsilon_r}(\epsilon_r - 1)$

➔ **P = $\boldsymbol{\epsilon_0(\epsilon_r - 1)E}$** $\quad because\ D = \frac{E}{\epsilon_0 \epsilon_r}$

Induced Dipole

Neutral atom



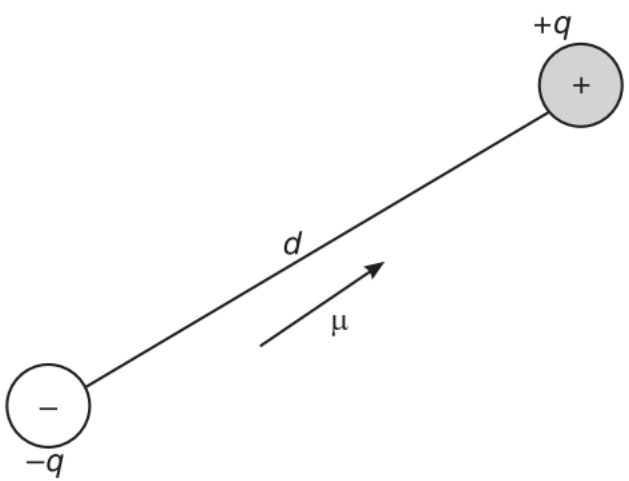


Atom in an electric field

$$q = Z\, e^{-}$$

$$\mu = q\, d$$

Dipole moment is a vector directed along the axis of the dipole from the negative charge to the positive charge.



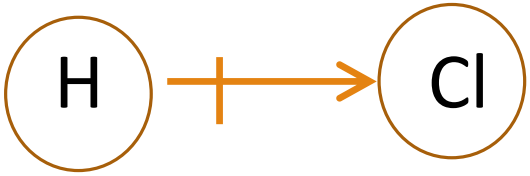
charges get displaced along the field direction due to E

Restoring force from Coulomb interaction try to pull them together

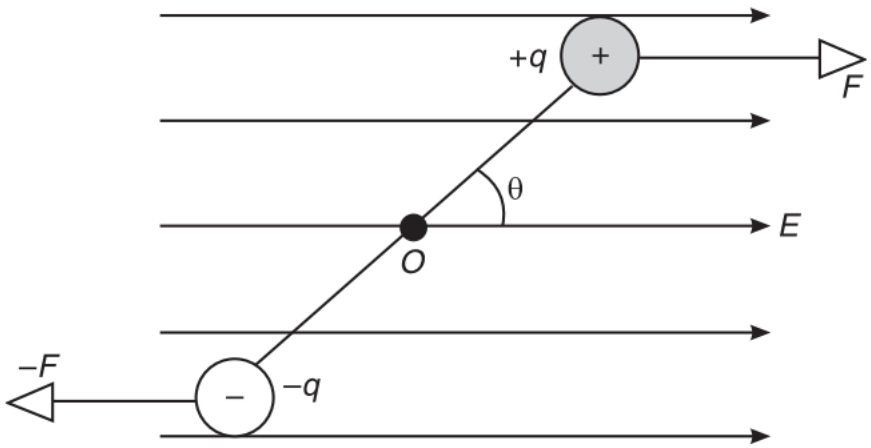
$$\mu_{induced} \propto E$$
$$\mu_{induced} = \alpha E$$

α is called polarizability of a material

Permanent Diploes

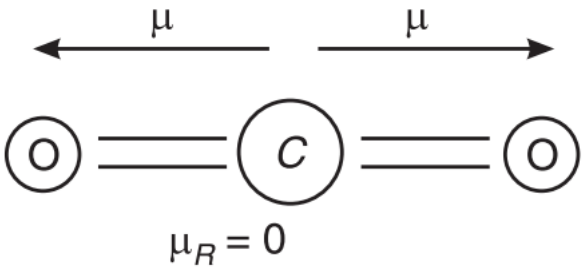


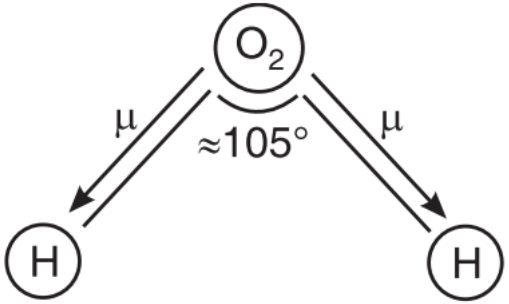
In HCl, Cl is electronegative and attracts the charge cloud towards it making a permanent dipole.



Torque on the dipole when placed in an electric field

$$\tau = qEsin\theta.d = \mu Esin\theta = \mu \times E$$

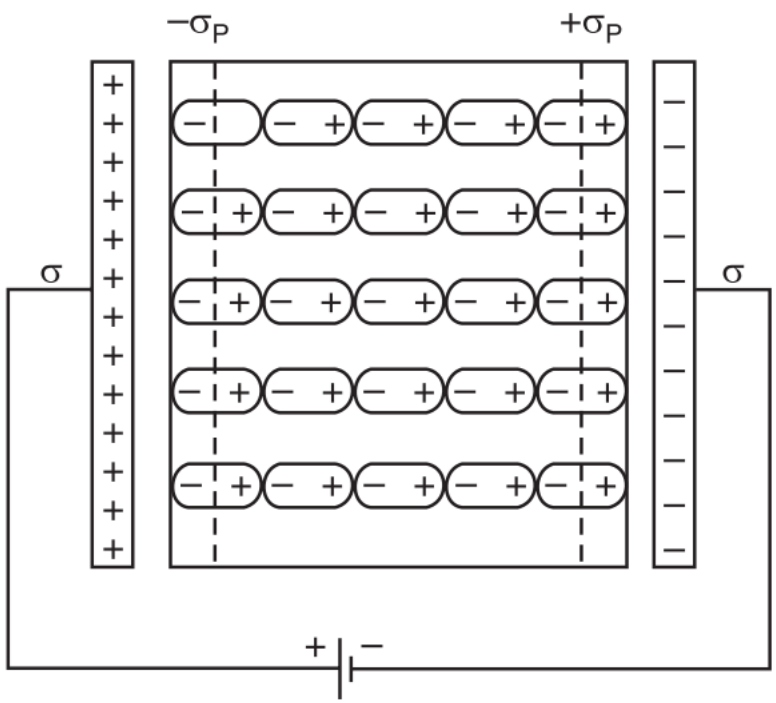




Non-polar dielectric: $\epsilon \approx 1 - 2.2$

Polar dielectric: $\epsilon \approx 3 - 8$

Dielectric Polarization



Non-polar dielectric material:

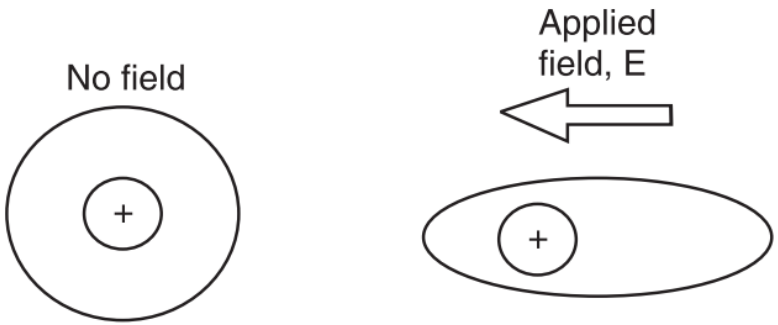
$P = N\mu = N\alpha E$

Polar dielectric material:

$P = N < \mu >$ $< \mu >$ is average alignment of the dipole moment

Different type of Polarization

Electronic Polarization



Occurs very fast: $10^{-14} – 10^{-15}$

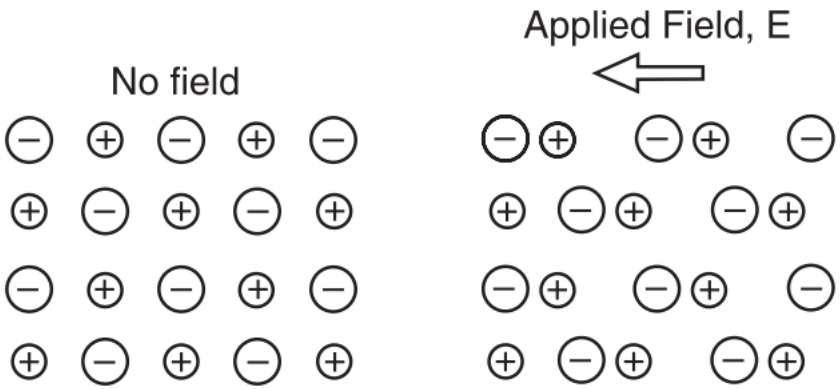
Electronic polarizability:

$$\alpha_e = 4\pi\epsilon_0 R^3$$

Dielectric constant:

$$\epsilon_r = 1 + \frac{N\alpha_e}{\epsilon_0}$$

Ionic Polarization



occurs in ionic crystals

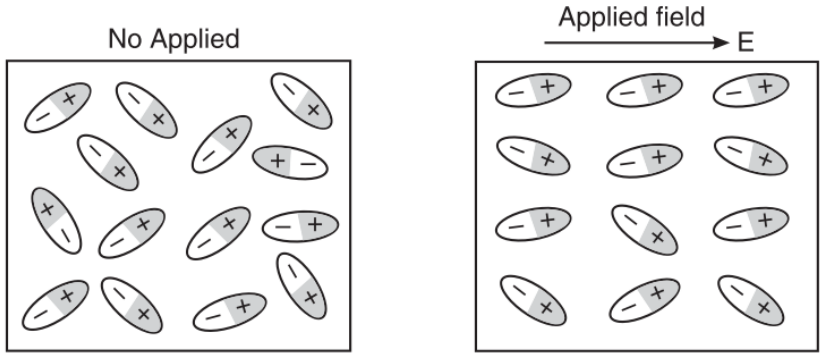
Takes longer time to build up

$10^{-11} – 10^{-14}$

Weaker than electronic polarization

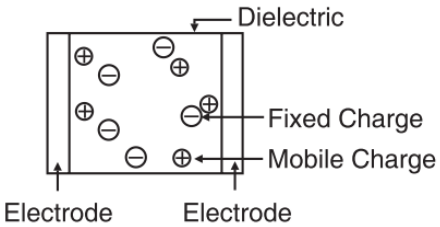
$$\alpha_i = \frac{\alpha_e}{10}$$

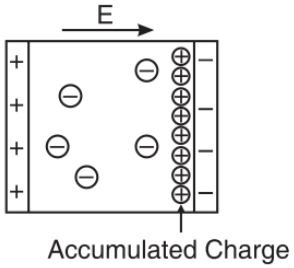
Orientation Polarization



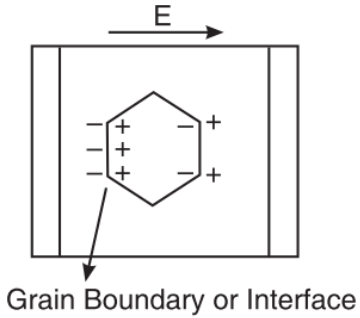
1. Occurs in gases, liquids and amorphous viscous substances
2. In solids, the molecules are fixed in their positions and their rotation is highly restricted by the lattice forces, small polarization (Water ~80, Ice~10)
3. Build up time 10^{-10} or more.

Space Charge Polarization





occurs in heterogeneous dielectric materials in which there is a change of electrical properties between different phases and in the homogeneous dielectrics that contain impurities, pores filled with air, inclusions of hygroscopic water etc



Numerical Problems

1. Calculate the polarization produced in dielectric medium of dielectric susceptibiltiy 6 when it is subjected to an electric field of 100 Vm^{-1}.

2. When NaCl crystal is subjected to an electric field of 50 V/cm, the resulting polarization is 2.215×10^{-7} C/m^2. Calculate relative permittivity of NaCl.